

UN SDG 11 — SUSTAINABLE CITIES & COMMUNITIES

Sensory-Friendly Urban Futures

Onboarding Discovery Presentation — wayfinding for sensory-sensitive commuters in Melbourne's CBD

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Agenda

- 1 Problem Statement
- 2 UNSDG Goal Linkage
- 3 Who: Persona & Empathy Map
- 4 Plan for the Iteration (MoSCoW)
- 5 High-Level Prototype
- 6 Open Data
- 7 Code Quality
- 8 Security Plan
- 9 Innovation

SECTION 01 / 07

Problem Statement

Elevator pitch, the problem, and why it matters

Sensory-sensitive adults are avoiding Melbourne's CBD, because no wayfinding tool accounts for sensory overload.

~20%

of people are **neurodivergent**
and cities remain designed for a single "average" user, with no sensory accessibility framework to guide planners

Source: UNSW City Futures Research Centre / Architecture & Design (2026)

High-Density Stressors

Construction, protests & policing spike sensory load in the CBD

A Policy Gap

Physical accessibility has standards. **Sensory accessibility has none**, planners are left relying on guesswork

Independent Access at Risk

Even Metro Trains has acknowledged the problem, launching the Hidden Disabilities Sunflower scheme to support commuters with sensory needs

How might we transform Melbourne's CBD into a naturally navigable environment for sensory-sensitive adults?

Linkage to UN SDG 11



**Sustainable Cities
& Communities**

Target 11.2 — Accessible Transit

Sensory-indicator routing lets commuters choose paths based on crowd density and stimulation level, not just speed, making independent CBD travel achievable for sensory-sensitive young adults.

Target 11.7 — Inclusive Public Spaces

By combining real-time alerts with recommendations for quiet, low-stimulation spaces, users can avoid overwhelming areas and easily find somewhere safe to reset within the CBD.

SECTION 02 / 07

Who

Persona walkthrough and empathy map analysis

Persona Walkthrough



Freddy, 28, Melbourne

- Neurodivergent marketer who commutes daily via train and tram
- Values independence, but finds crowds, noise, and construction overwhelming
- Mentally maps backup routes before every trip, which is exhausting
- Wears a Hidden Disabilities Sunflower lanyard, though his disability isn't visible

"I don't need the fastest route. I need the route that helps me arrive calm, focused, and ready for my day."

Frustrations

- No single, reliable source of sensory-aware information, forcing him to check multiple apps before every trip
- Unpredictable pedestrian congestion and last-minute route changes leave him constantly on edge

Goals

- Travelling independently through the CBD without relying on others for support
- Arriving calm and ready for his day, rather than anxious and mentally drained from pre-planning

Empathy Map Analysis

SAYS

He always has a backup route mapped out in his head before he even leaves the house.

THINKS

“What if the platform is packed again and I have to push through the crowd?”

DOES

Checks 2–3 different apps every morning and picks quieter carriages or off-peak trains when he can.

FEELS

Feels anxious and on edge during unpredictable, crowded commutes — relieved when a route goes exactly as planned.

SECTION 03 / 07

Plan for the Iteration

MoSCoW priorities and the high-level prototype

Iteration Scope and Prioritisation

Objective: Help sensory-sensitive commuters compare route conditions and choose a lower-congestion route in Melbourne CBD.

	MUST HAVE	SHOULD HAVE	COULD HAVE	WON'T HAVE THIS ITERATION
EPIC 1 Sensory-Aware Route Planning <i>(Choose a calmer route)</i>	US 1.1 Sensory Indicators (Compare route comfort) US 1.2 Avoid Congested Corridors (Reduce crowd exposure)	—	US 1.3 Personal Crowd Threshold (Prototype only)	—
EPIC 2 Sensory Environment Monitoring <i>(Find support when overwhelmed)</i>	—	US 2.1 Sensory Refuge Locations (Stretch goal)	—	US 2.2 One-Hour Predictive Alerts (Future backlog)
COMMITTED BUILD US 1.1 + US 1.2 Stretch US 2.1 · Prototype only US 1.3 · Future backlog US 2.2				

Current LeanKit Kanban

NOT STARTED

BACKLOG - FUTURE WORK

Epic 2.0

Sensory Environment Monitoring

0 OF 2

Epic 2.0 US 2.1

Sensory refuge locations

0 OF 1

US 2.1 AC 2.1.1

Display nearby sensory refuge locations

US 2.1 AC 2.1.2

Show refuge name and location type

US 2.1 AC 2.1.3

Show no-results message

EPIC 2.0 US 2.2

One-hour prediction

STARTED

TO DO THIS ITERATION

Epic 1 US 1.2

Avoid congested corridors

0 OF 4

US 1.2 AC 1.2.1

Identify highly congested route segments

US 1.2 AC 1.2.2

Recommend a lower-congestion route

US 1.2 AC 1.2.3

Identify the comparatively lower-congestion route

DOING

IN PROGRESS

Epic 1

Sensory-Aware Route Planning & Real-Time Navigation

0 OF 4

Epic 1 US 1.1

User Story 1.1 Sensory indicator

0 OF 4

US 1.1 AC 1.1.1

Display High/Low Sensory labels

US 1.1 AC 1.1.2

Communicate indicator using text, not colour alone

US 1.1 AC 1.1.3

Show sensory information unavailable state

BLOCKED

DONE

TASK 1.1 A

Figma: Design sensory indicator states

TASK 1.3 A

Figma: Prototype crowd-sensitivity preference

TASK QA 01

Verify prototype flow and navigation

US 1.3 AC 1.3.1 (PRO...

Apply selected crowd-sensitivity preference

TASK 1.2 A

Figma: Design congestion route states

TASK 2.1 A

Figma: Design sensory refuge states

0 OF 3

EPIC 1 US 1.3

Personal crowd threshold

0 OF 2

US 1.3 AC 1.3.2 (PRO...

Demonstrate alternative-route recommendation

100%

Acceptance Criteria Card Demonstration

Details

Tasks

Parent Cards

1

Child Cards

Dependencies

Comments

Attachments

History and Health

PARENT CARD

Epic 1 US 1.1

User Story 1.1 Sensory indicator

0 OF 4

US 1.1 AC 1.1.2

Communicate indicator using text, not colour alone

Title

Communicate indicator using text, not colour alone

Description

H1H2BBIUSAA

Given a sensory indicator is displayed,
When the user views the route,
Then the indicator must be communicated using text and not colour alone.

Card Type

Documentation

Size

Tags

Save

Cancel

Assigned Members

Type to search...

Planned Start

Planned Finish

Header

US 1.1 AC 1.1.2

Priority

Blocked

Lane

TO DO THIS ITERATION

Parent Card

User Story 1.1 Sensory indicator

External Link

http://www.example.com

AgilePlace

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SECTION 03 · HIGH-LEVEL PROTOTYPE

Prototype Journey — A Calmer Route

Preference → route comparison → optional refuge support

01 PERSONALISE & PLAN

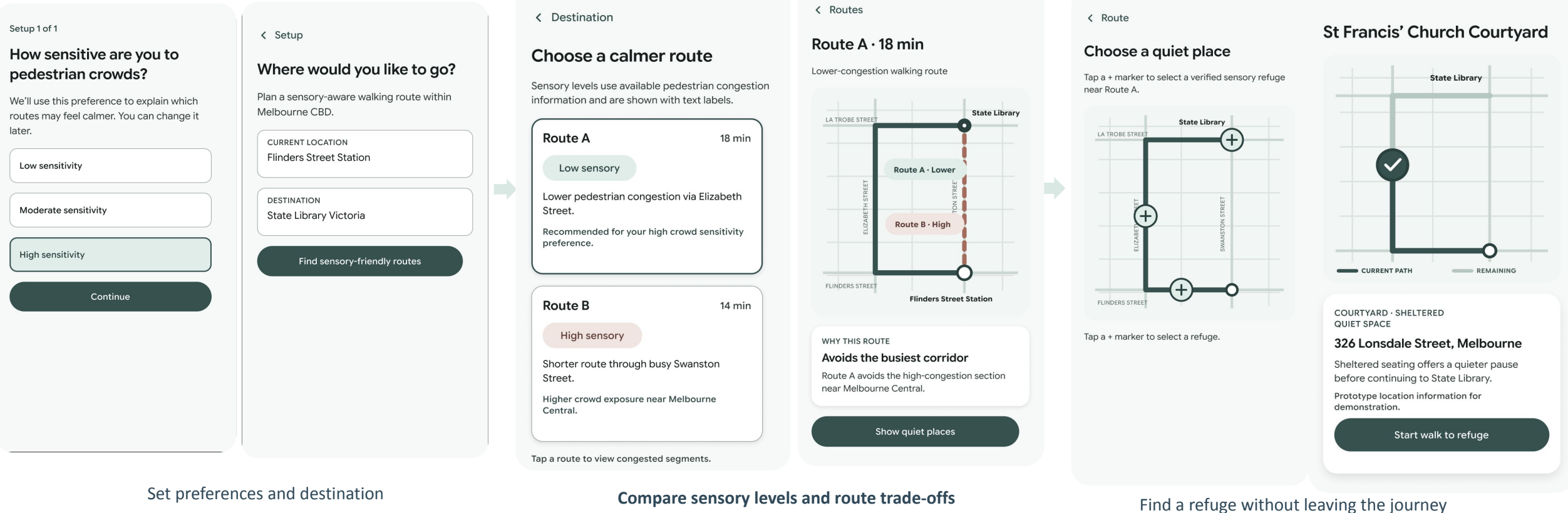
US 1.3 · Prototype only

02 COMPARE & CHOOSE

US 1.1 + US 1.2 · Must Have

03 FIND SUPPORT

US 2.1 · Stretch



Set preferences and destination

Compare sensory levels and route trade-offs

Find a refuge without leaving the journey

Committed flow: clear sensory labels + an explainable lower-congestion route recommendation.

Figma: <https://www.figma.com/design/3tnjTlvoB6c93kXvp9KP/CalmPath-Prototype-Demonstration?node-id=0-1&t=7WW9loqDc3b7mTNx-1>

Edge Cases — Clear, Honest and Recoverable

When information or options are limited, the interface explains what changed and preserves user control.

01 DATA UNAVAILABLE

< Destination

Choose a calmer route

Sensory levels use available pedestrian congestion information and are shown with text labels.

Route A18 min

Low sensory

Lower pedestrian congestion via Elizabeth Street.

Recommended for your high crowd sensitivity preference.

Route B14 min

Sensory information unavailable

Pedestrian data is currently unavailable for this route.

No sensory recommendation is shown.

Tap a route to view congested segments.

No false certainty
Missing data never becomes a guessed sensory label.

02 ALL ROUTES CONGESTED

< Destination

Routes have some congestion

Compare the available routes and choose the option with comparatively lower pedestrian congestion.

All available routes currently contain some congestion.

Route A18 min

Moderate sensory

Comparatively lower congestion than Route B.

Recommended because it currently has comparatively lower pedestrian congestion than Route B.

Route B14 min

High sensory

Higher pedestrian congestion.

All available routes contain some congestion.

Route A is recommended for comparison; it is not congestion-free.

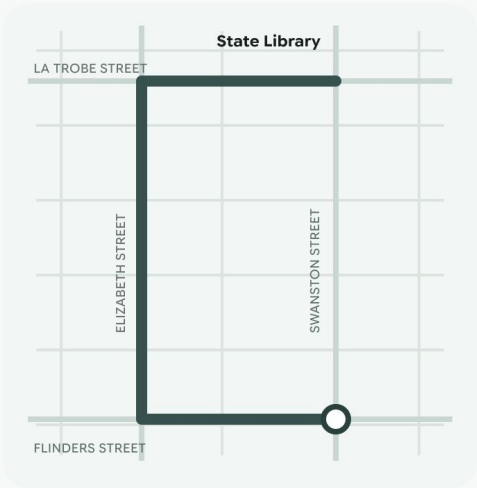
Honest comparison
Recommends comparatively lower congestion — not a congestion-free route.

03 NO REFUGE NEARBY

< Route

No quiet places nearby

No nearby sensory refuge locations found.



Try expanding the search area or continue on your selected route.

Return to route

Recoverable action
Explains the empty result and returns the user to the selected route.

Design outcome: limitations remain visible, recommendations remain accurate, and every state offers a next step.

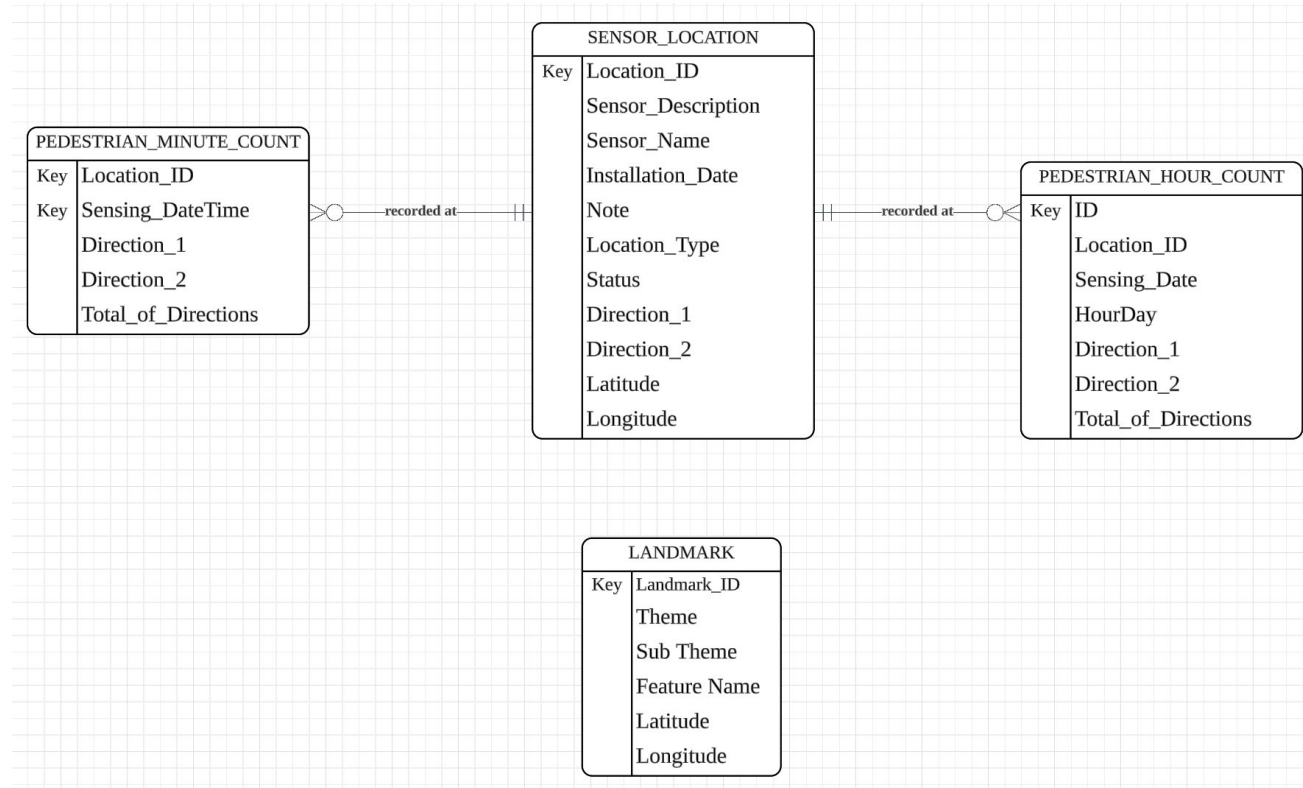
SECTION 04 / 07

Open Data

Datasets, ERD, and the insights they unlock

Datasets We're Using

- 1 Pedestrian Counting System – Past Hour (Counts per Minute) — melbourne.vic.gov.au
- 2 Pedestrian Counting System – Sensor Locations datasets — melbourne.vic.gov.au
- 3 Pedestrian Counting System – Past Hour (Counts per Hour) tasetes — melbourne.vic.gov.au
- 4 Landmarks and places of interest, including schools, theatres, health services, sports facilities, places of worship, galleries and museums. — melbourne.vic.gov.au



Insights

Hindsight

- Historical pedestrian counts by hour and weekday shows predictable commuter and weekend patterns
- Landmark type shows which categories of places are historically linked to highest pedestrian traffic

Foresight

- Combining crowd stability with nearby landmark identify calmer locations user can retreat to
- A personalized crowd risk score, weighted by the users own sensitivity to noise, will help tailor ‘busy’ and ‘quiet’ spaces to what actually matters to them

Insight

- A model using time of day, weekday and recent trend data predicts next-hour crowd levels
- Predictive alerts can flag specific streets or area that is likely to become crowded soon

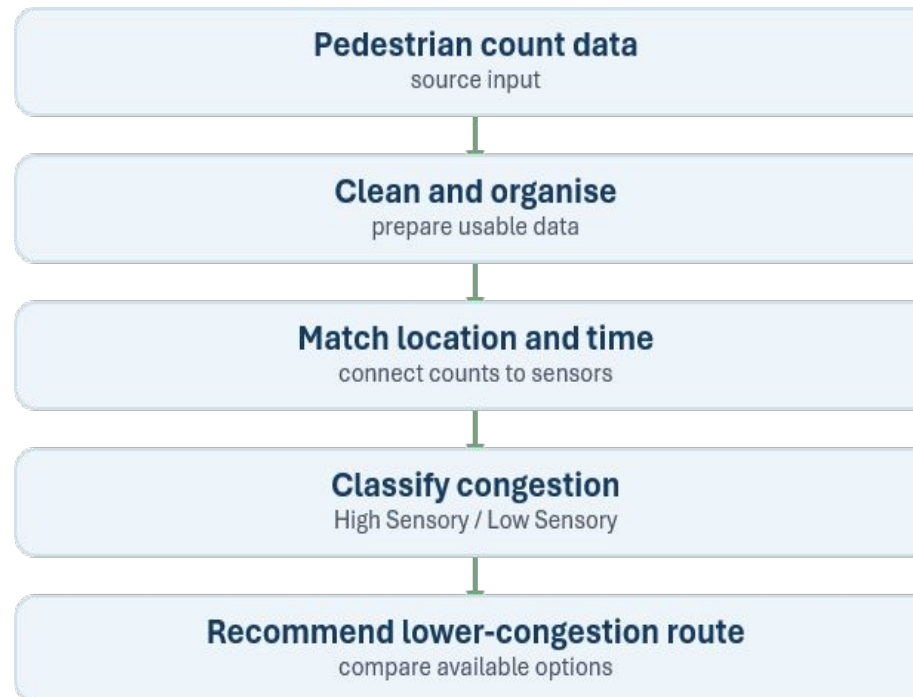
Rule-Based Classification and Workflow

Current onboarding iteration: transparent congestion classification, not predictive AI.

CURRENT APPROACH

- No machine-learning or AI model is used in this iteration.
- A transparent rule-based method classifies congestion as **High Sensory** or **Low Sensory**.
- The method is explainable and easy to verify during onboarding.
- It directly supports the current Must Have user stories for congestion identification and route comparison.

WORKFLOW



USER BENEFIT

Freddy gets a route option that is easier to understand and less likely to feel overwhelming.

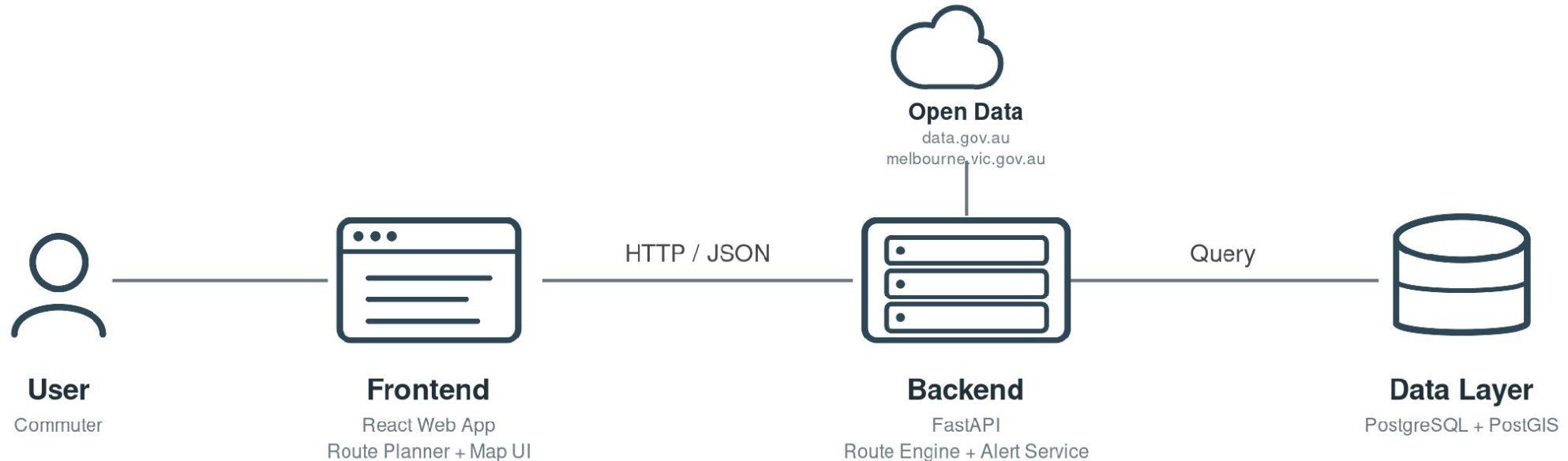
Future work: Predictive modelling for US 2.2 may be explored in a later iteration, but it is not included in the current onboarding iteration.

SECTION 05 / 07

Code Quality

System architecture and how we keep the codebase healthy

System Architecture & Quality Assurance



- Linting on every commit: ESLint/Prettier (React), Black/Flake8
- Peer review required before merge — feature-branch Git workflow
- Automated tests: Jest (frontend), Pytest (FastAPI backend)
- CI (GitHub Actions) blocks merge on lint/test failure
- Type checking: mypy (Python), TypeScript/PropTypes (React)
- API self-documented via FastAPI OpenAPI/Swagger

Definition of Done ties quality to release: critical/high defects resolved and accessibility-tested before mentor sign-off.

SECTION 06 / 07

Security Plan

Security considerations, QA, and penetration testing

Security & Quality Assurance

Quality Assurance of Code

- Security linters (Bandit, ESLint) + dependency scanning on every commit
- Security checklist on every peer review; tests cover input validation & permissions
- CI blocks merge on high-severity findings

Security Considerations

- API & database security: Input validation, rate limiting, and parameterised queries to prevent injection attacks
- Data exposure: No sensitive details exposed in error messages or logs
- Encryption: HTTPS/TLS enforced for all client-server data transmission
- Access control: Backend restricted to only the database access it needs
- Privacy: Location and sensory data treated as personal information (Australian Privacy Act), kept only as long as needed
- Fairness: Sensory data never used to disadvantage users (Disability Discrimination Act)

Penetration Testing

- Scope: API endpoints, any authentication/session handling, input validation on destination/route entry
- Timing: Conducted before major milestone demo/final submission
- Method: Basic vulnerability scan (OWASP ZAP) + manual check for the high-risk items from the risk table
- Documentation: Findings and remediation logged and reported back to team/mentors

SECTION 07 / 07

Innovation

The standout feature — and why it matters to our user

What Makes This Innovative?

THE INNOVATIVE FEATURE

Explainable Sensory-Aware Route Comparison

- Combines pedestrian counts, sensor locations, and landmarks into a unified data structure.
- Uses transparent rule-based classification to label routes as High or Low Sensory.
- Compares routes by pedestrian congestion, not just travel time.
- Explains route recommendations, highlighting streets or areas with lower crowd exposure.

BENEFIT TO THE USER

- Helps sensory-sensitive commuters choose routes that match their comfort level.
- Reduces exposure to crowded and overwhelming areas.
- Simplifies complex pedestrian data with clear sensory labels.
- Supports more confident and independent travel across Melbourne CBD.
- Provides a foundation for personalised preferences and future predictive alerts.

Thank You

Questions & Discussion

LeanKit: <https://monashie.leankit.com/board/2494529218>

Project Governance: https://drive.google.com/drive/folders/1kDCI_DaLfpKXQ2MsPiMn6piflPXW-0qN